Signed Interpolation for Partial Relational Tasks

This walkthrough explains why the signed interpolation $(1-\gamma)X - \gamma I$ enables dual decoders to solve both feature and position tasks simultaneously, while standard interpolation $(1-\gamma)X + \gamma I$ fails.

The Problem: Entangled Information

For a task requiring both feature and position prediction, we need the attention output to preserve both types of information. Consider the standard adaptable bottleneck:

$$V = \big[(1-\gamma)X_c + \gamma I\big]W_v$$

At intermediate γ values (e.g., $\gamma = 0.5$), the value matrix becomes:

$$V = 0.5X_c + 0.5I$$

Example ($L = 3$): Context item at position 1 with feature 2:

$$X_c[\text{pos } 1] = \begin{bmatrix}0 & 0 & 1\end{bmatrix}$$

After value interpolation ($\gamma = 0.5$):

$$V[\text{pos } 1] = 0.5\begin{bmatrix}0 & 0 & 1\end{bmatrix} + 0.5\begin{bmatrix}1 & 0 & 0\end{bmatrix} = \begin{bmatrix}0.5 & 0 & 0.5\end{bmatrix}$$

Problem: Feature info (index 2) and position info (index 0) are now *mixed* in the same vector. Linear decoders cannot separate them:

- Feature decoder W_f needs to extract "feature = 2" from $[0.5, 0, 0.5]$
- Position decoder W_p needs to extract "position = 0" from the *same* vector $[0.5, 0, 0.5]$

There exists no linear projection that can simultaneously recover both pieces of information from an entangled representation.

The Solution: Signed Interpolation

Change the sign on the identity term:

$$V = \big[(1-\gamma)X_c - \gamma I\big]W_v$$

At $\gamma = 0.5$:

$$V = 0.5X_c - 0.5I$$

Same example: Position 1, feature 2:

$$V[\text{pos } 1] = 0.5\begin{bmatrix}0 & 0 & 1\end{bmatrix} - 0.5\begin{bmatrix}1 & 0 & 0\end{bmatrix} = \begin{bmatrix}-0.5 & 0 & 0.5\end{bmatrix}$$

Key insight: The opposite signs create *separable* components:

- **Positive component** ($+0.5$ at index 2): feature information
- **Negative component** (-0.5 at index 0): position information

How Dual Decoders Extract Information

The dual decoders learn to separate the signed components:

Feature Decoder Strategy

W_f learns to pick out *positive* activations (feature signal from X_c):

$$W_f \approx \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{(identity-like, captures positive part)}$$

$$\text{logits}_f = V \cdot W_f = \begin{bmatrix} -0.5 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.5 & 0 & \mathbf{0.5} \end{bmatrix}$$

$\arg\max = 2$ ✓ (correct feature)

Position Decoder Strategy

W_p learns to pick out *negative* activations (position signal from $-I$):

$$W_p \approx \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{(negative identity, flips sign)}$$

$$\text{logits}_p = V \cdot W_p = \begin{bmatrix} -0.5 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} \mathbf{0.5} & 0 & -0.5 \end{bmatrix}$$

$\arg\max = 0$ ✓ (correct position)

General Principle: Differential Encoding

The signed interpolation creates a **differential encoding**:

$$V[\text{pos } i,\ \text{feature } j] = (1-\gamma) \cdot e_j - \gamma \cdot e_i$$

where e_k is the k-th basis vector. This is analogous to:

- **Signal processing**: In-phase vs. quadrature components (I/Q modulation)
- **Neuroscience**: ON vs. OFF cells in retinal encoding
- **Linear algebra**: Encoding information in orthogonal subspaces

The opposite polarities act as *tags* that make the information linearly separable, enabling simple linear decoders to extract both signals from the same vector.

Conclusion: Standard interpolation $(1-\gamma)X + \gamma I$ entangles feature and position information at intermediate γ values. The signed variant $(1-\gamma)X - \gamma I$ uses opposite signs to create separable components that dual linear decoders can extract independently.